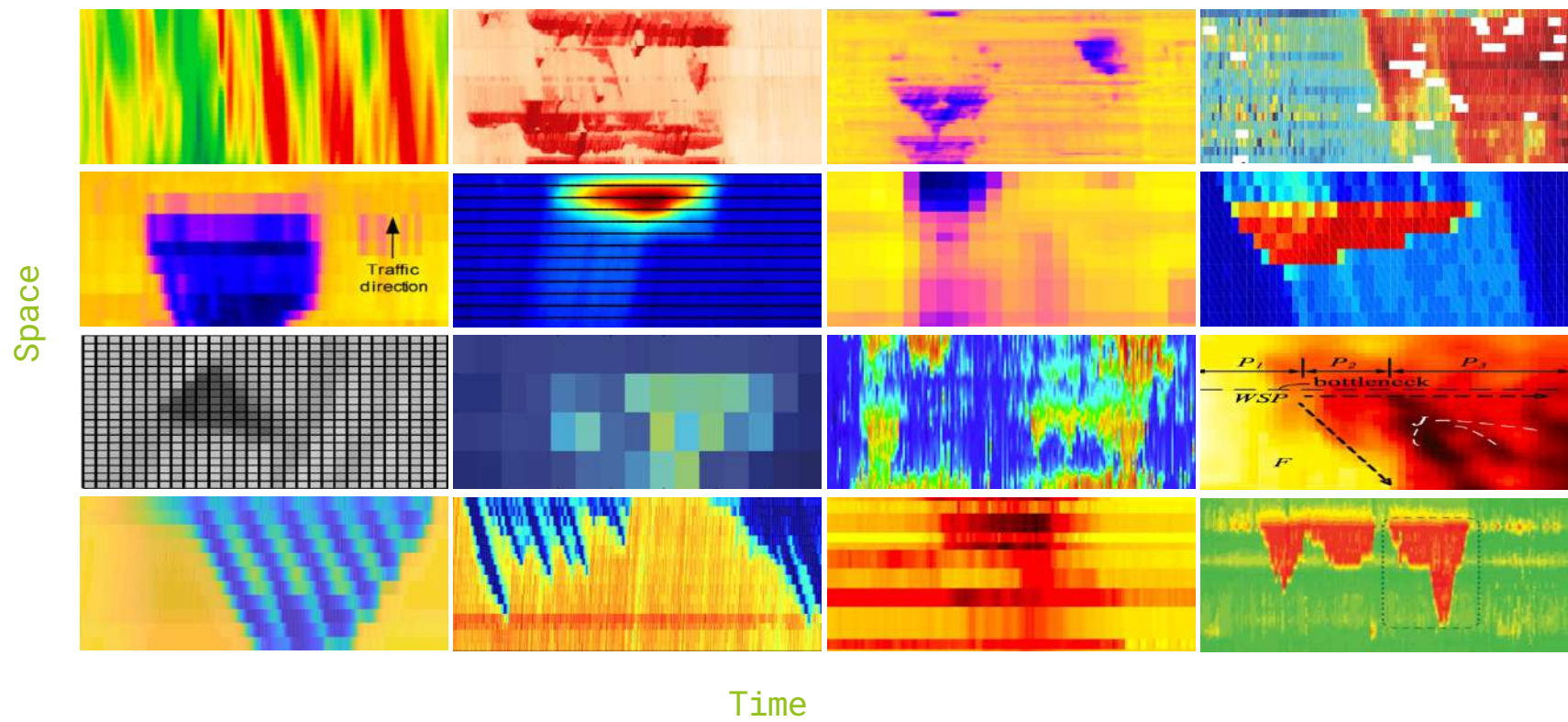


Beautifying Traffic via Enhancing Time-space Traffic Diagrams

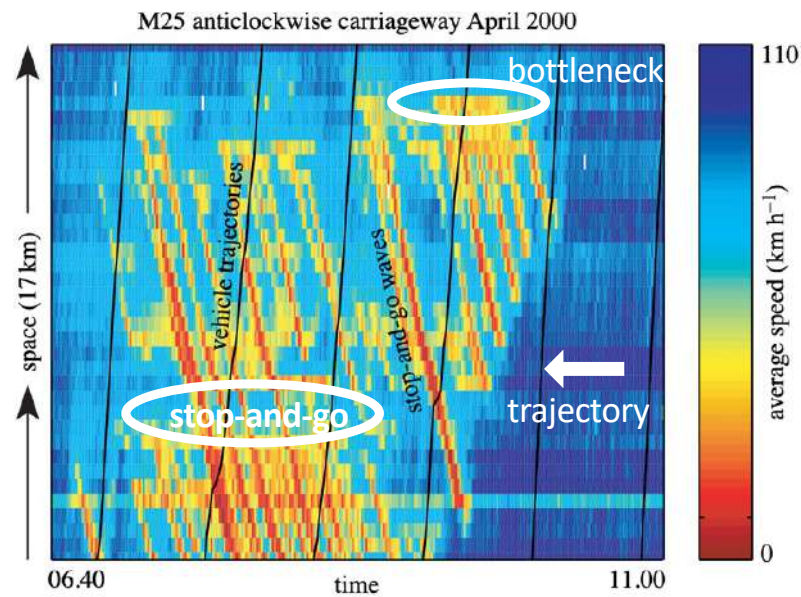
Zhengbing He

www.GoTrafficGo.com

Time-space (TS) Diagram



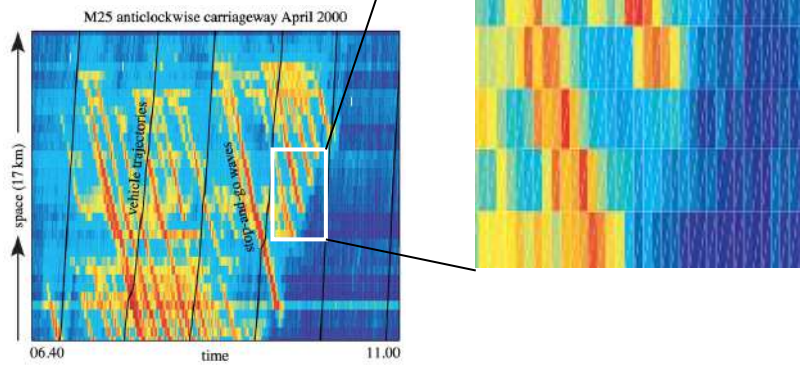
Time-space (TS) Diagram



*Wilson, R. E. (2008). Mechanisms for spatio-temporal pattern formation in highway traffic models. *Philosophical Transactions. Series A*, 366(1872), 2017–2032.

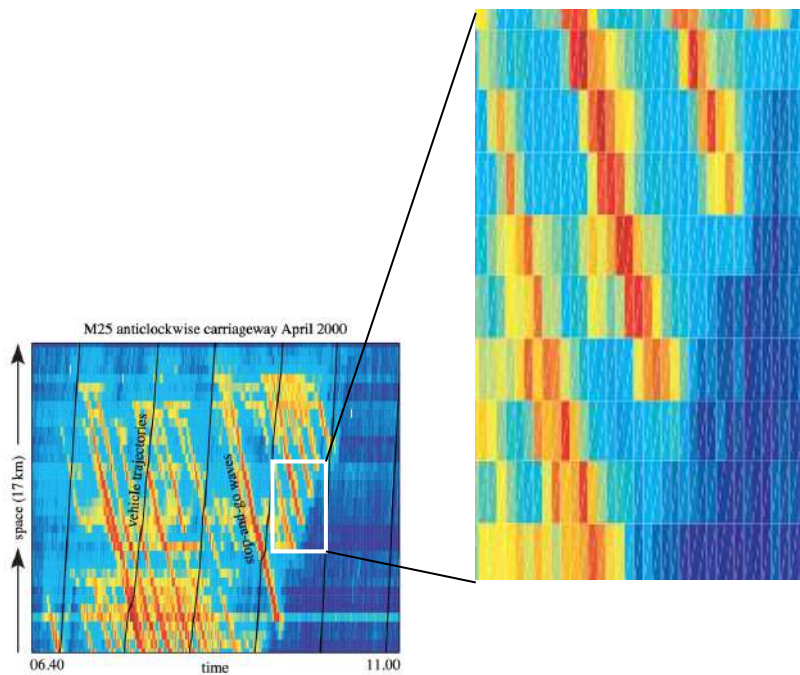
Time-space (TS) Diagram

toothed shape | error

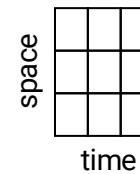


Time-space (TS) Diagram

toothed shape | error



Cause



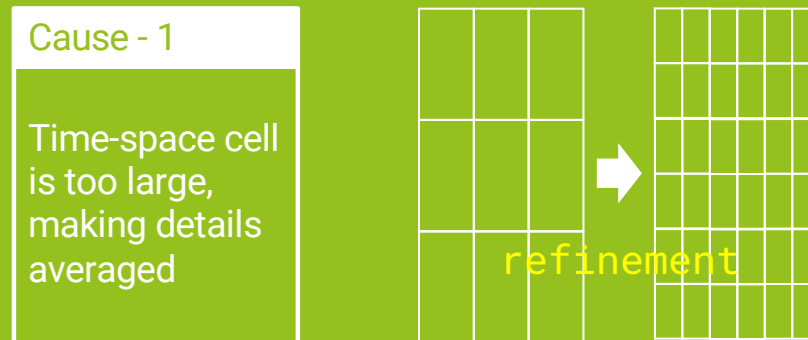
Cause - 1

Time-space cell is too large, making details averaged

Cause - 2

Rectangular cells cannot incorporate the propagation direction of traffic waves

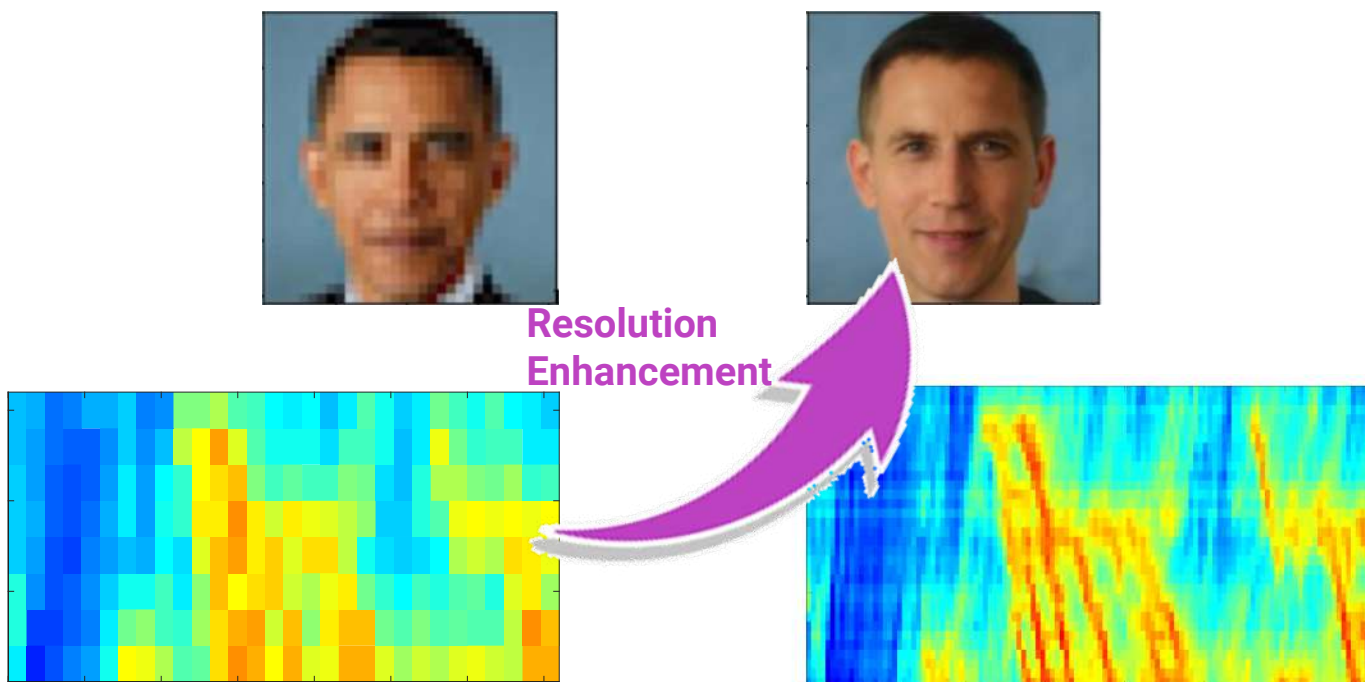
Refining Time-space Diagrams



Reference

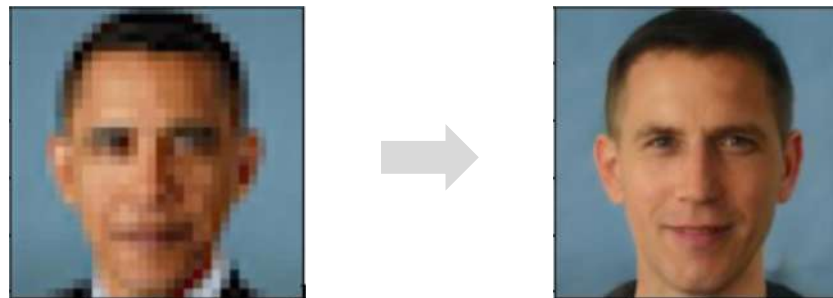
- Zhengbing He, Refining time-space traffic diagrams: A simple multiple linear regression model, *IEEE Transactions on Intelligent Transportation Systems*, 10.1109/TITS.2023.3316593

Refining Time-space Diagrams



Refining Time-space Diagrams

Super-resolution: enhancing the resolution of an imaging system



Deep-learning based Super-resolution Approaches

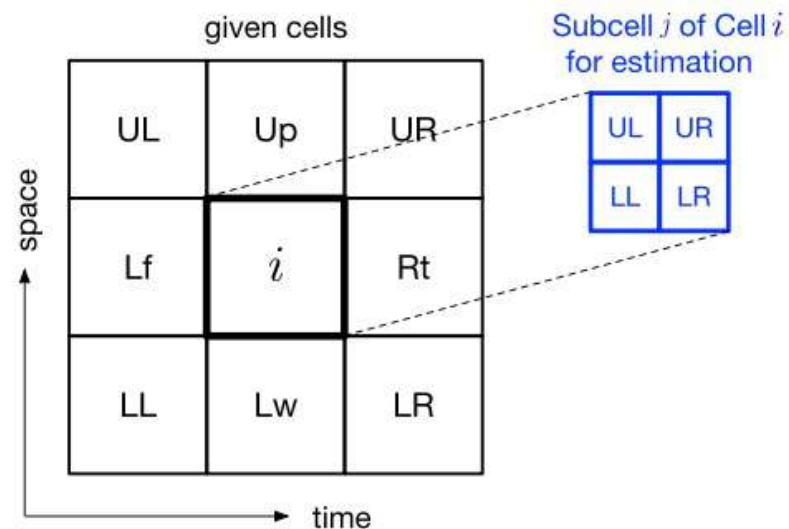
- high computational cost
- Requirement of a large number of high-fidelity images to train models

main obstacle

Refining Time-space Diagrams

Definition

Given a TS diagram, partition **each cell** (i) into **four homogenous sub-cells** (j) and estimate the traffic speed of the four sub-cells to achieve the goal of refining the TS diagram by enhancing its resolution



Refining Time-space Diagrams

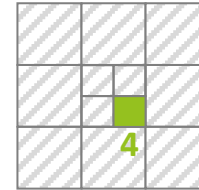
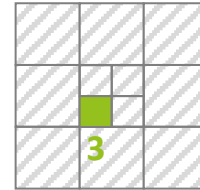
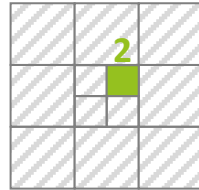
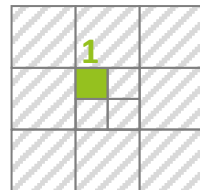
Solution

- **Step 1:** Determine the traffic condition of each cell based on a threshold of traffic speed
- **Step 2:** For each sub-cell, we estimate its speed by fitting the following multi-linear regression model

$$\hat{y}_{ik}^j = \mathbf{p}_k^j \cdot \mathbf{x}_i + \varepsilon_k^j$$

where

- $\mathbf{x}_i = [x_i, x_{LL}, x_{LW}, x_{LR}, x_{Rt}, x_{UR}, x_{UP}, x_{UL}, x_{Lf}]$ -- ground-true speeds of cell i and its surrounding cells
- $\mathbf{p}_k^j = [p_k^j, p_{kLL}^j, p_{kLW}^j, p_{kLR}^j, p_{kRt}^j, p_{kUR}^j, p_{kUP}^j, p_{kUL}^j, p_{kLf}^j]$ -- regression coefficient
- ε_k^j -- a random residual term

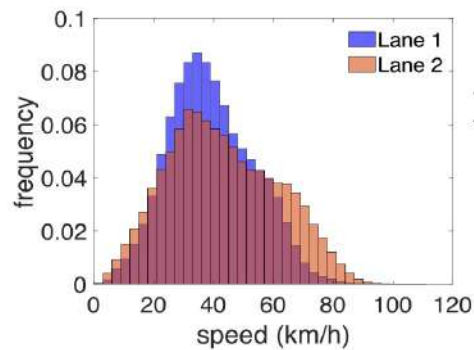
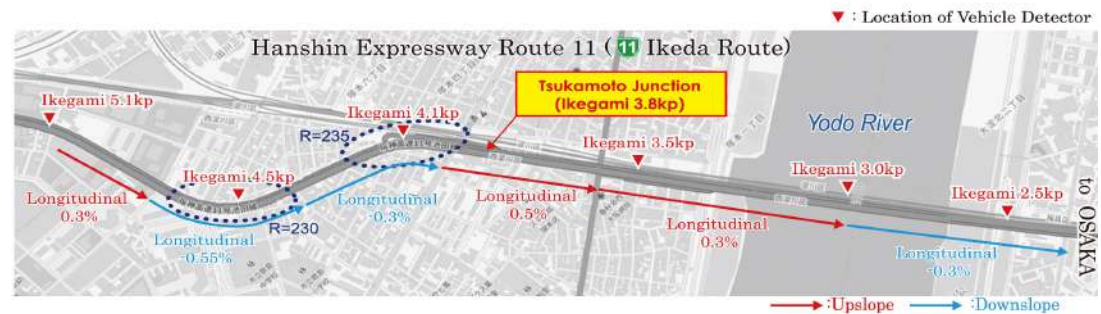


Refining Time-space Diagrams

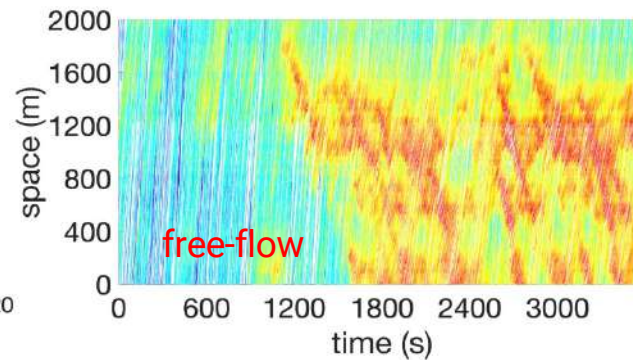
Data for model fitting

ZenTraffic Dataset

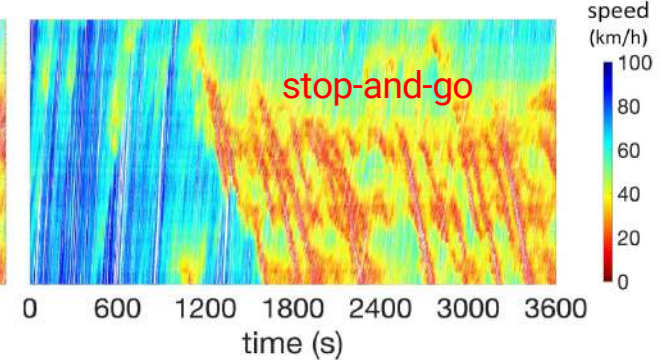
- Ikeda Route 11 in Japan
- 2017
- 2-lane freeway
- 2 km segment
- 1st hour of total 5 hours



(a) Comparison of traffic speed



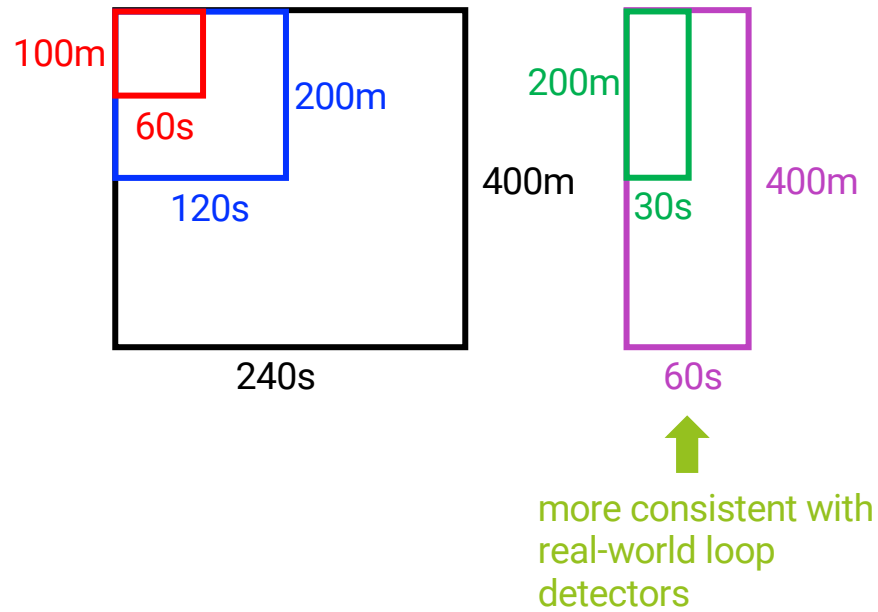
(b) Vehicle trajectories in Lane-1



(c) Vehicle trajectories in Lane-2

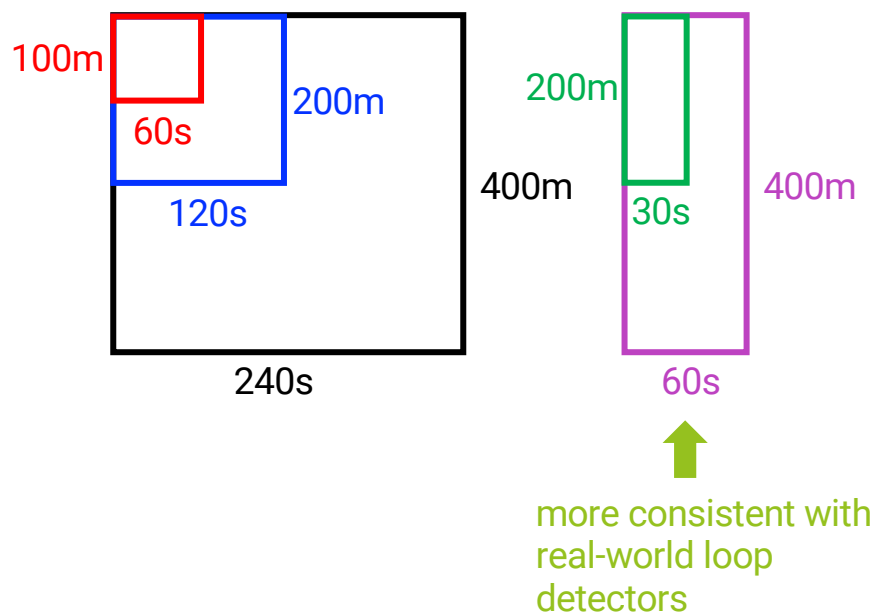
Refining Time-space Diagrams

Results



Refining Time-space Diagrams

Results



Cell size	Subcell				Regression coefficients of Equation 1												R ²																																																																																																																																																																																																																																																																																																																																							
	size	k	j	#	P _L ^j	P _{LL} ^j	P _{LLw} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j		P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j	P _{LR} ^j	P _{LRi} ^j

Refining Time-space Diagrams

Validation

	Dataset	Segment length (km)	Collection duration (h)	Lane #	Country
Model fitting	Ikeda-Route-11	2	1 (F001)	2	Japan
Evaluation	Ikeda-Route-11	2	4 (F002-F005)	2	Japan
	Wangan-Route-4	1.6	5 (F001-F005)	2	Japan
	US-101	0.64	0.75 (7:50-8:35)	5	United States
	Total	4.24	9.75	9	

different time

ZenTraffic

different location

ZenTraffic

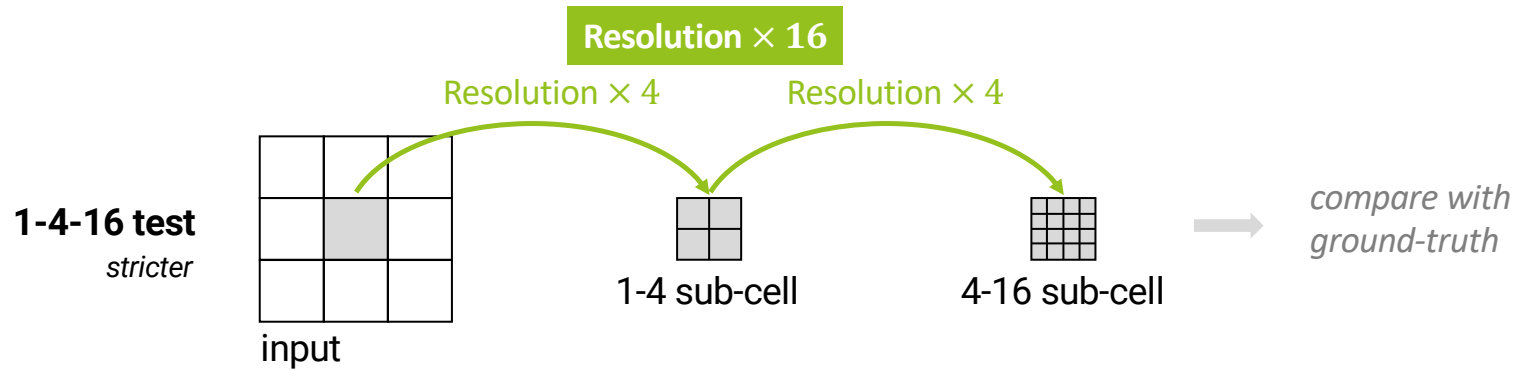
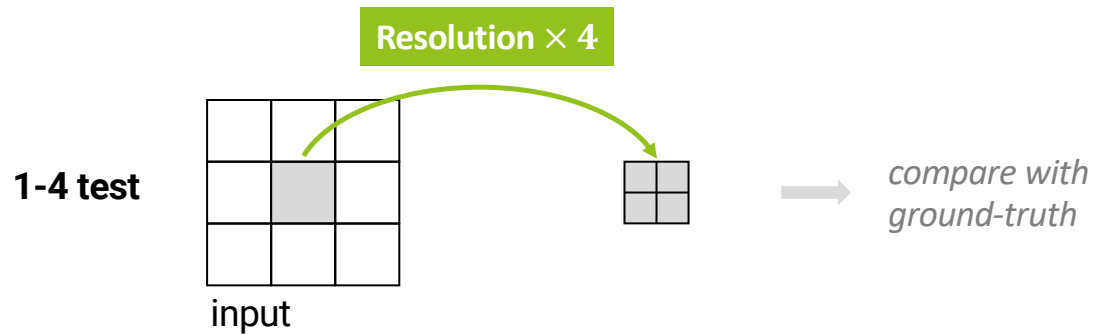
different country

NGSIM

TRANSFERABILITY

Refining Time-space Diagrams

Tests



Refining Time-space Diagrams

Results

RESULTS OF MODEL EVALUATION: 1-4 TEST

Cell size	1-4 subcell size	j	Ikeda-Route-11 (F002-F005)			Wangan-Route-4 (F001-F005)			US-101		
			Subcell #	MAE	MAPE	Subcell #	MAE	MAPE	Subcell #	MAE	MAPE
30s	15s	LL	38028	1.915	0.040	36941	1.603	0.032	4894	2.158	0.086
×	×	LR	38035	1.911	0.040	36948	1.601	0.032	4894	2.157	0.085
50m	25m	UR	38025	1.904	0.040	36946	1.609	0.032	4894	2.144	0.080
		UL	38025	1.919	0.041	36947	1.611	0.032	4894	2.211	0.085
60s	30s	LL	9153	2.292	0.050	8790	1.911	0.039	1100	2.823	0.098
×	×	LR	9153	2.267	0.050	8790	1.929	0.040	1100	2.748	0.093
100m	50m	UR	9153	2.229	0.049	8790	1.925	0.039	1100	2.754	0.095
		UL	9153	2.301	0.049	8790	1.947	0.039	1100	2.927	0.099
120s	60s	LL	2088	2.783	0.061	1974	2.228	0.046	—	—	—
×	×	LR	2088	2.781	0.063	1974	2.242	0.047	—	—	—
200m	100m	UR	2088	2.660	0.061	1974	2.243	0.044	—	—	—
		UL	2088	2.782	0.061	1974	2.232	0.044	—	—	—
240	120s	LL	416	3.316	0.073	390	2.916	0.058	—	—	—
×	×	LR	416	4.431	0.085	390	4.543	0.084	—	—	—
400m	200m	UR	416	3.634	0.077	390	3.419	0.062	—	—	—
		UL	416	4.434	0.085	390	4.298	0.079	—	—	—
30s	15s	LL	8789	2.132	0.049	8344	1.715	0.036	—	—	—
×	×	LR	8790	2.221	0.050	8344	1.864	0.038	—	—	—
200m	100m	UR	8791	2.059	0.047	8344	1.781	0.035	—	—	—
		UL	8789	2.182	0.048	8344	1.902	0.037	—	—	—
60s	30s	LL	1925	2.836	0.064	1758	2.260	0.045	—	—	—
×	×	LR	1925	2.712	0.060	1758	2.361	0.047	—	—	—
400m	200m	UR	1925	2.678	0.063	1758	2.512	0.048	—	—	—
		UL	1925	2.640	0.059	1758	2.418	0.046	—	—	—

1-4 test

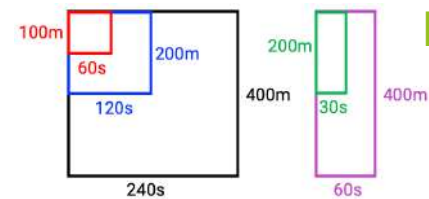
MAE: (1.6, 5) km/h MAPE: (0.03, 0.1)

RESULTS OF MODEL EVALUATION: 1-4-16 TEST

Cell size	1-4-16 subcell size	j	Ikeda-Route-11 (F002-F005)			Wangan-Route-4 (F001-F005)			US-101		
			Subcell #	MAE	MAPE	Subcell #	MAE	MAPE	Subcell #	MAE	MAPE
60s	15s	LL	31994	3.195	0.070	29893	2.606	0.054	3006	3.804	0.160
×	×	LR	31987	3.218	0.073	29899	2.658	0.055	3006	3.787	0.158
100m	25m	UR	31987	3.218	0.073	29897	2.605	0.053	3006	3.783	0.153
		UL	31990	3.240	0.072	29897	2.611	0.053	3006	3.894	0.155
120s	30s	LL	6272	3.953	0.093	5440	3.130	0.065	—	—	—
×	×	LR	6272	3.941	0.097	5440	3.110	0.066	—	—	—
200m	50m	UR	6272	3.813	0.091	5440	2.923	0.061	—	—	—
		UL	6272	3.894	0.090	5440	3.058	0.063	—	—	—
240s	60s	LL	768	5.808	0.132	480	4.002	0.085	—	—	—
×	×	LR	768	7.720	0.160	480	6.134	0.122	—	—	—
400m	100m	UR	768	6.254	0.137	480	4.776	0.098	—	—	—
		UL	768	7.988	0.166	480	6.093	0.120	—	—	—
60s	15s	LL	3732	4.401	0.109	2300	3.300	0.071	—	—	—
×	×	LR	3735	4.373	0.109	2300	3.210	0.070	—	—	—
400m	100m	UR	3735	4.371	0.109	2300	3.225	0.071	—	—	—
		UL	3734	4.251	0.105	2300	3.155	0.070	—	—	—

1-4-16 test

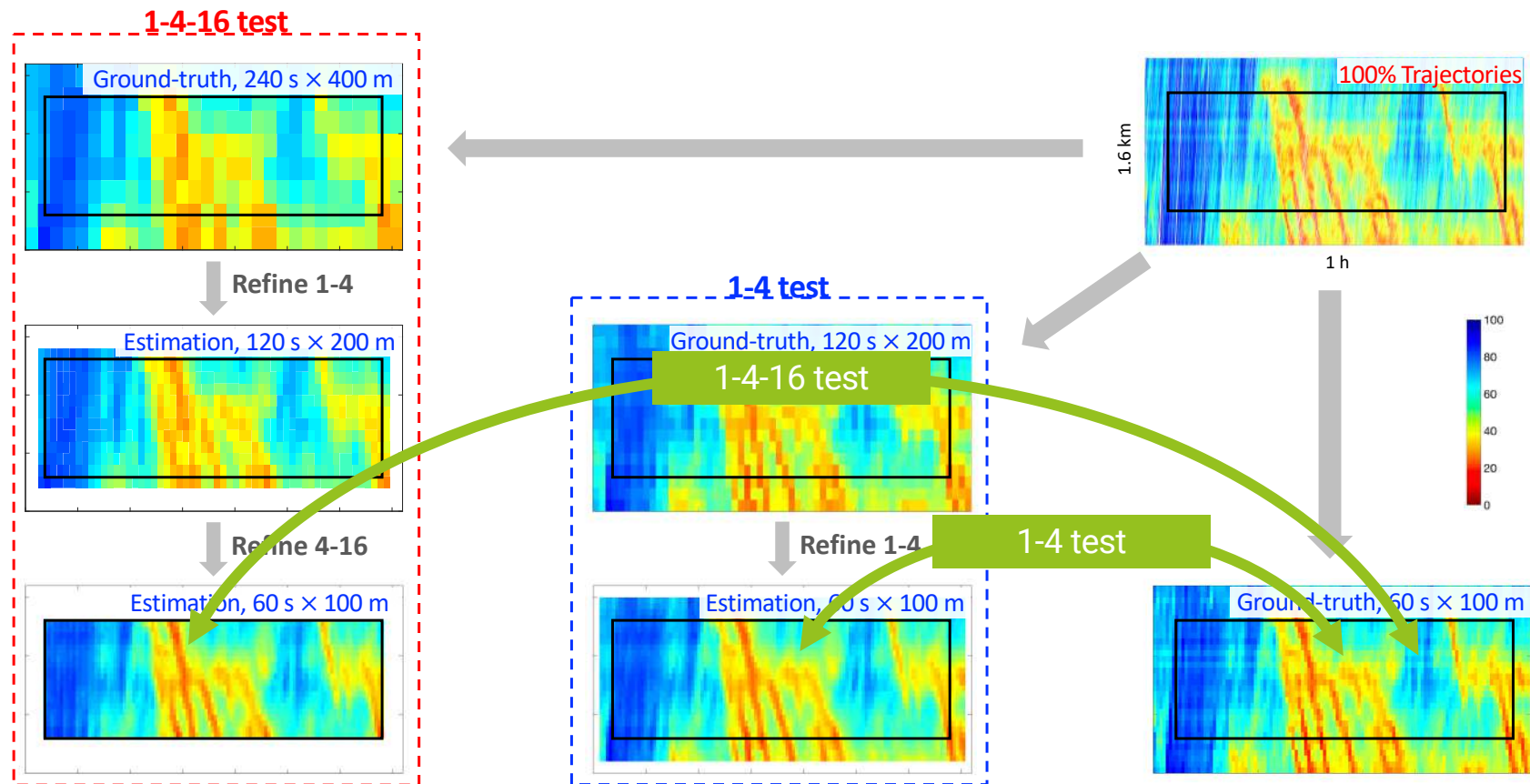
MAE: (2.5, 8) km/h MAPE: (0.05, 0.17)



Ideal results !!

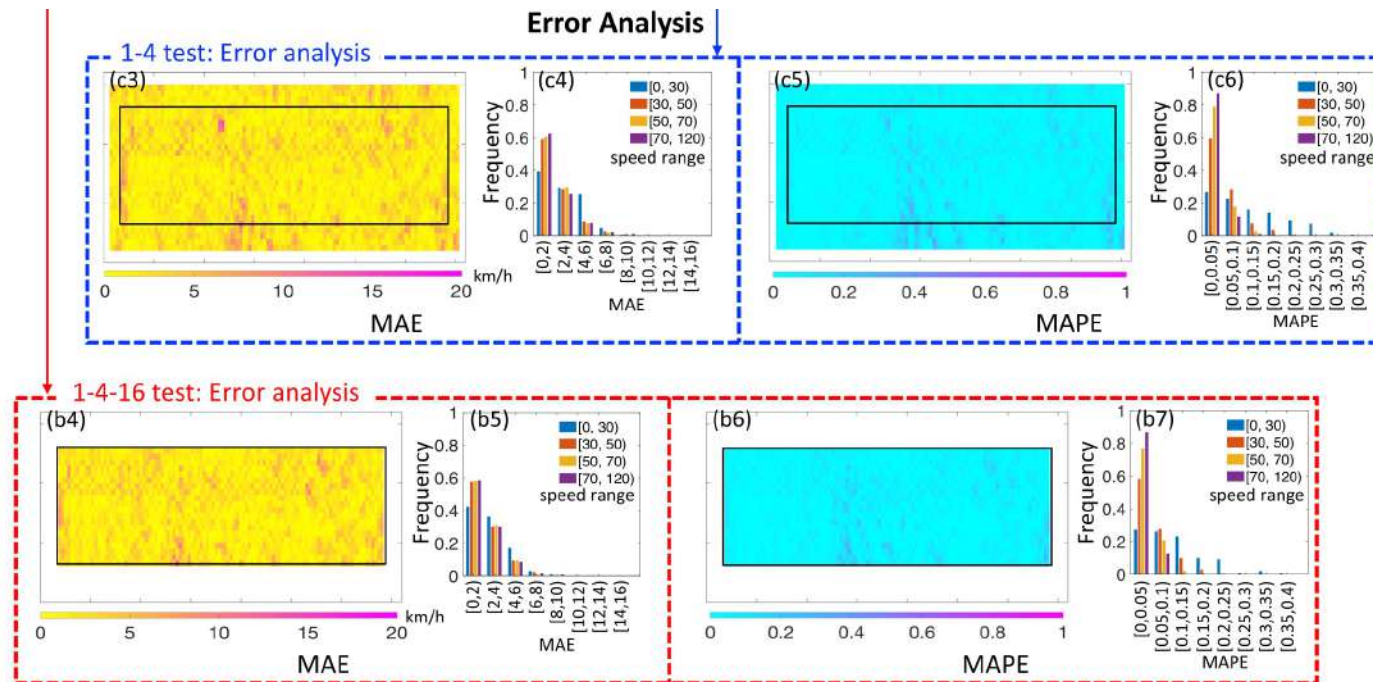


Refining Time-space Diagrams



Refining Time-space Diagrams

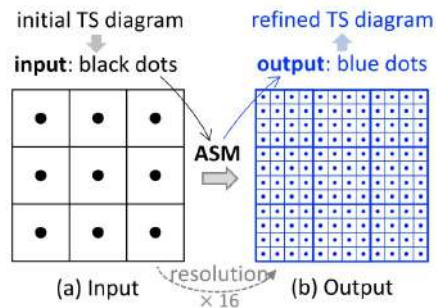
Error Analysis: Errors are independent of traffic states and traffic patterns



Refining Time-space Diagrams

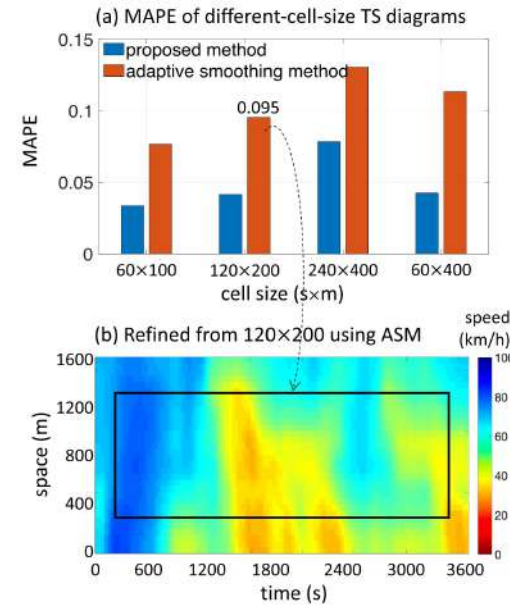
Comparison with the classic adaptive smoothing method* (ASM)

model settings



Parameter	Value
smoothing width in time coordinate	$\sigma = 60$ s
smoothing width in space coordinate	$\tau = 100$ m
propagation speed of perturbations in free traffic	$c_{\text{free}} = 70$ km/h
propagation speed of perturbations in congested traffic	$c_{\text{cong}} = -15$ km/h
crossover from congested to free traffic	$V_{\text{thr}} = 60$ km/h
transition width between congested and free traffic	$\Delta V = 20$ km/h

results

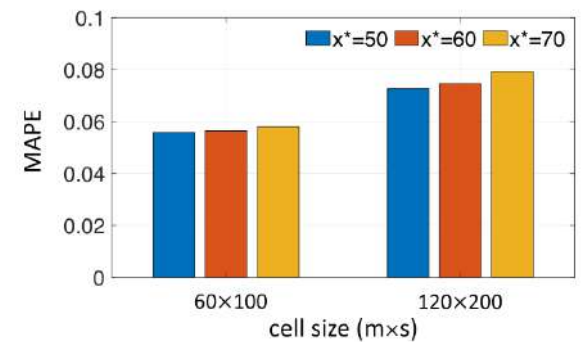


* Treiber, Helbing, Reconstructing the spatiotemporal traffic dynamics from stationary detector data, Cooperat. Transp. Dyn., 2002

Refining Time-space Diagrams

Discussion

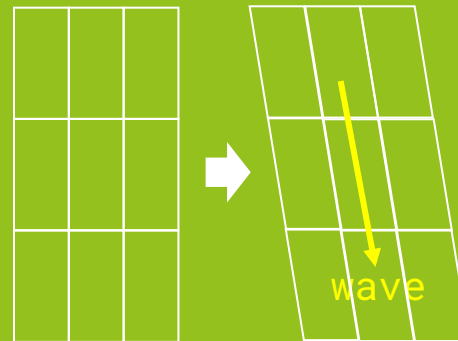
- Sensitivity of the Speed Threshold
- Necessity of Considering Traffic Conditions
- Transferability of the Method
 - Data from different times/locations/countries
 - Abundant traffic patterns have been involved in the evaluation
 - Similarity of traffic characteristic in general
- Simpleness of the Method
- Feasibility of Describing Nonlinear Traffic Using Linear Methods
- Multicollinearity of Regression
- Computational Cost



New Way of Construction

Cause - 2

Rectangular cells cannot incorporate the propagation direction of traffic waves

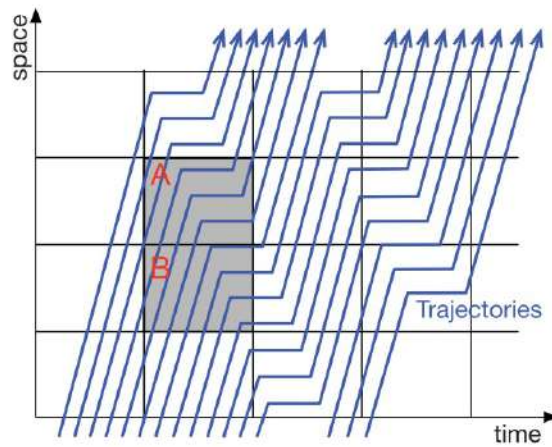


Reference

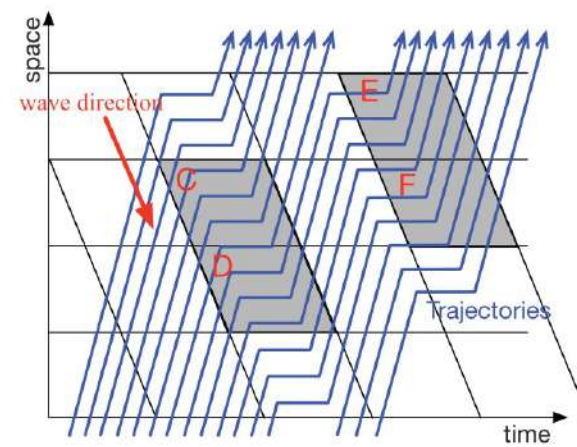
- Zhengbing He, et al. Constructing spatiotemporal speed contour diagrams: using rectangular or non-rectangular parallelogram cells? *Transportmetrica B*, 7(1), 44–60. 2019

New Way of Construction

Basic Idea



rectangular parallelogram (RP)

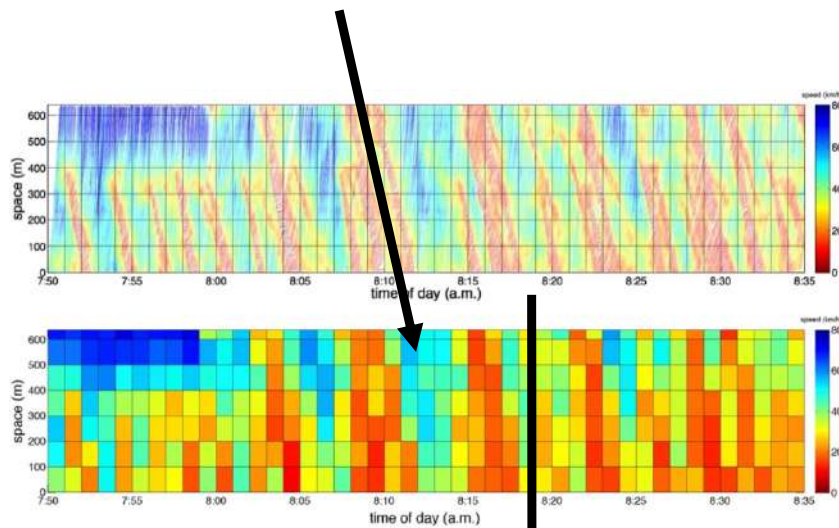


non-rectangular parallelogram (nRP)

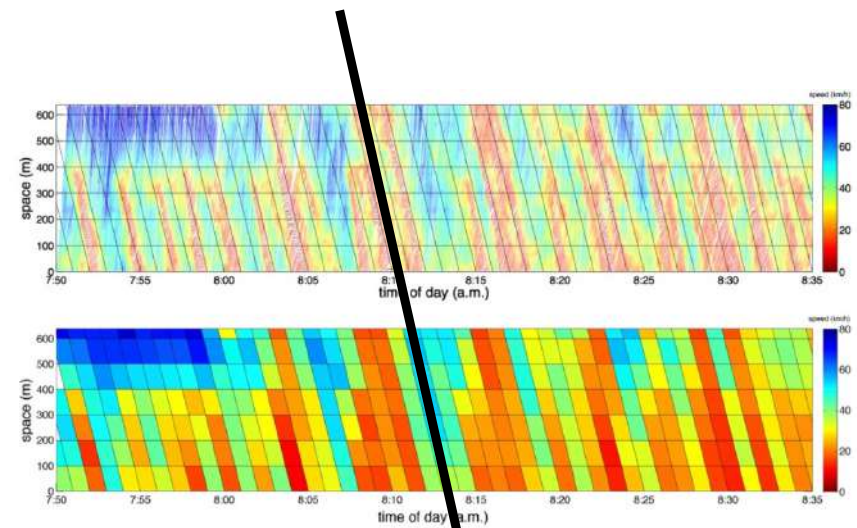
with wave direction as slope

New Way of Construction

Visual Comparison



rectangular parallelogram (RP)

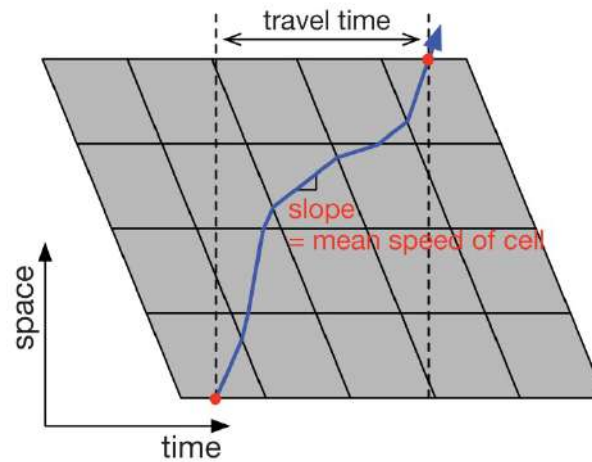


non-rectangular parallelogram (nRP)

with wave direction as slope

New Way of Construction

Metric of Comparing TS Diagrams



New Way of Construction

RP vs. nRP Comparison

- given Complete Trajectory Data
- given Partial Probe Data

Sensitivity Analysis of Wave Speed Estimations

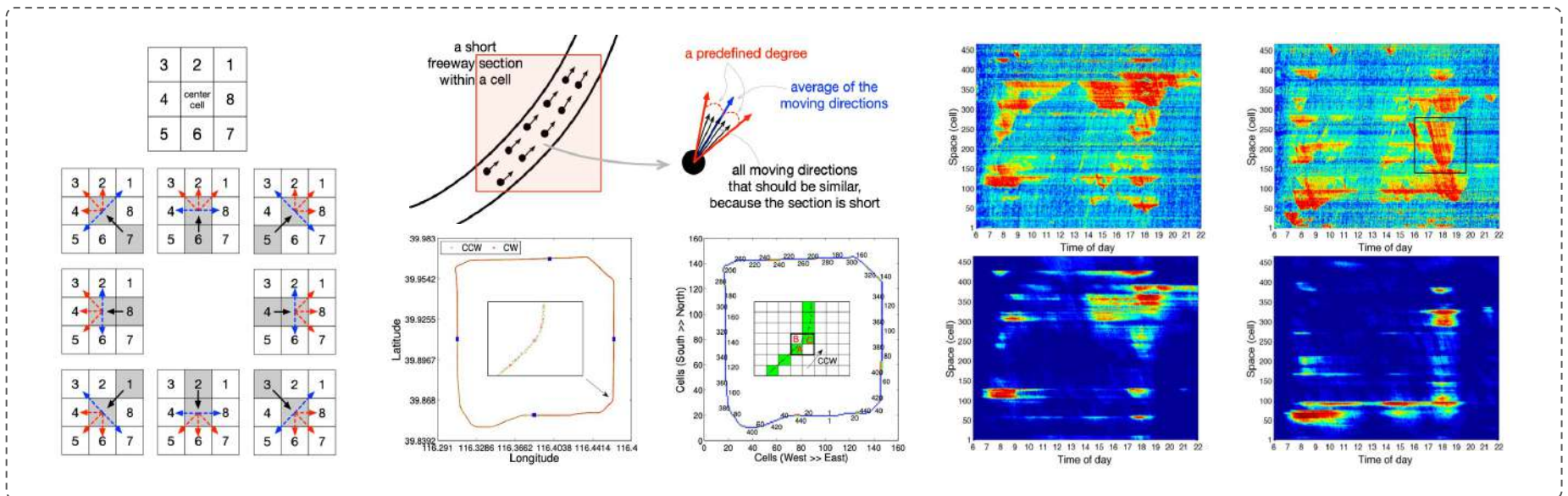
- Globally, wave speed $\in [-20, -10]$ km/h
- nRP is better, when setting wave speed at -20 km/h and -10 km/h

Selected Studies based on Time-space Diagram

Congestion Sensing

Solely Floating Car Data: Freeway

- Mapping-to-Cell: No need of map-matching
- Computationally efficient: Discretization of two-dimensional space
- Traffic dynamics on Freeway models

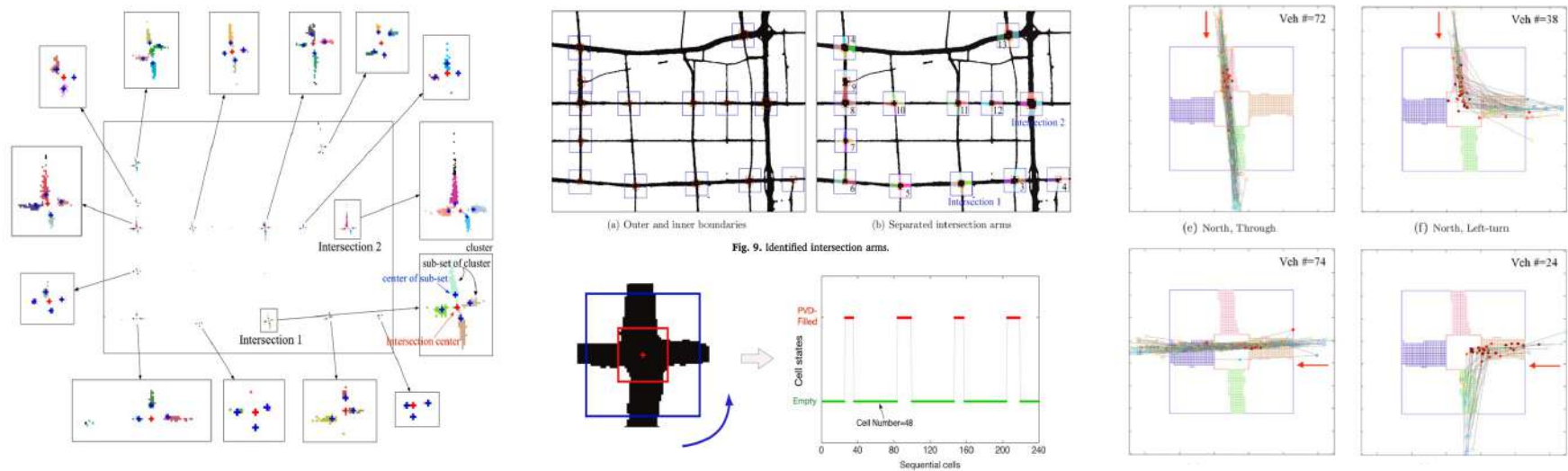


Zhengbing He, et al. Mapping to Cells: A Simple Method to Extract Traffic Dynamics from Probe Vehicle Data. *Computer-Aided Civil and Infrastructure Engineering*, 32(3), 252–267. 2017.

Congestion Sensing

Solely Floating Car Data: Urban Road

- Mapping-to-Cell: No need of map-matching
- Computationally efficient: Discretization of two-dimensional space
- Screening intersections for turn-level congestion in a road network

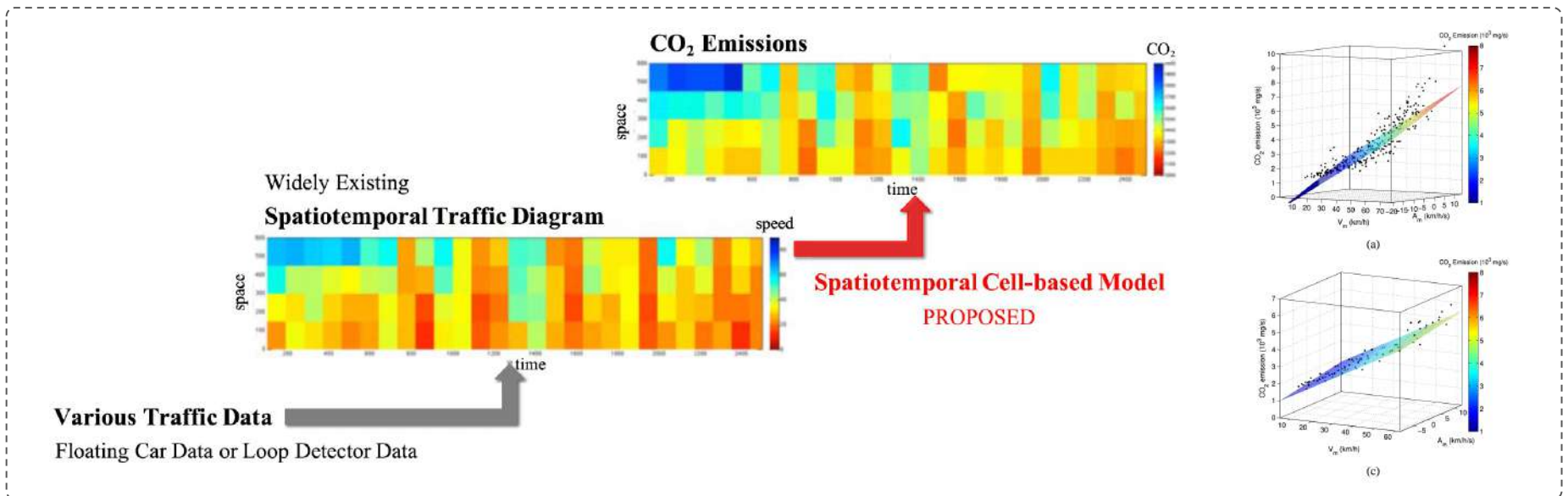


Zhengbing He, et al., *Network-wide Identification of Turn-level Intersection Congestion Solely Using Sparse Probe Vehicle Data*. *Transportation Research Part C*, 108(October), 320–339, 2019.

Mesoscopic CO₂ Emission Estimation Model

Estimate CO₂ Emission from Time-space Diagram

- Traffic dynamics is considered in the estimation
- First emission estimation model based on 100% traffic flow data
- Vast potential for applications



Zhengbing He, et al. Estimating Carbon Dioxide Emissions of Freeway Traffic: A Spatiotemporal Cell-Based Model. *IEEE Transactions on Intelligent Transportation Systems*, 21(5), 1976–1986, 2020.

Thank You



Website
www.GoTrafficGo.com